

SHADI OSSAMA

MCP & Plug-In Connectors Expert · LLM Power User

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Egypt · Languages: Arabic (native) · English (fluent) · Available 20+ hrs/week, remote

SUMMARY

Daily LLM power user who runs **three MCP servers in production**, driving real work — 3D modelling, game engines, desktop software, research — through connected tools rather than chat alone. Maintains a **128-page knowledge base with 40 decision records** on what AI tooling got right, what it got wrong, and why. Builds evaluation systems as a habit: a three-stage blind-peer-review pipeline for comparing model answers, failure taxonomies distilled from hundreds of logged errors, and a context-degradation canary that flags declining output quality before it shows.

MCP & CONNECTOR EXPERIENCE

- Operate **three MCP servers in daily production use** — **blender-mcp**, **robloxstudio-mcp (54 tools)**, and Roblox's official Studio server — driving every 3D-modelling and game-engine change through tool calls instead of manual editing across **46 documented build sessions**.
- Eliminated a recurring class of desync bugs by establishing a standing rule that engine work goes through MCP tool calls and never disk-only edits, after diagnosing that files written to disk silently fail to reach the running editor.
- Maintain a **documented tool-evaluation registry** recording every MCP server, plugin, and repository assessed as installed or declined *with the reason* — including **two rejections on security grounds**: a swarm framework requiring curl-pipe-to-shell installation with full-permission plugins, and an AI gateway proxying OAuth in violation of provider terms.
- Authored **five custom Claude Code skills** that load domain rules, verify MCP connectivity, and enforce project conventions at session start, plus **hooks** that block writes to immutable source directories and auto-commit the workspace at session end.
- Run **personal-life workflows** on connected tools — a **career-search pipeline** that scrapes job boards, auto-dedupes against a running application tracker, and triages employer replies through **Gmail**, plus a **scheduled** daily briefing and **browser** and **desktop automation** for research and verification.

AI EVALUATION, RUBRICS & FAILURE ANALYSIS

- Run a **three-stage anti-sycophancy evaluation pipeline** — independent agents answer blind, blind-peer-review and **rank** each other, then a chairman synthesises the vetted answer — with an opt-in **cross-vendor mode comparing GPT, Gemini, Claude and Grok** through OpenRouter.
- Analysed **91 AI-generation failures across 17 builds** and distilled them into **8 recurring error patterns and 10 standing rules** — a working rubric for judging generated output — eliminating repeat failures in every session that followed.
- Documented a **five-stage context-degradation taxonomy** — instruction omission → convention drift → unfounded assumptions → fabricated file paths → session failure — and instrumented it with a canary token that drops first, giving an early warning before output quality visibly degrades.
- Caught **three classes of false measurement** where the instrument rather than the system was wrong: a sandbox reporting a successful system write **three times** for a task that never existed, a network benchmark whose transport mismatch **inverted both conclusions**, and a contrast checker that read gradient fills as transparent and reported failures at 1.05:1 for text measuring 7:1.
- Wrote **40 decision records** capturing the option chosen, the options rejected, and what breaks if reversed, plus **per-session debug logs** cataloguing every error with root cause and fix across two projects and the **107 standing rules** derived from them.

TECHNICAL PROJECTS

Exile Codex — Python + WebView2 desktop application

2026

Solo build · github.com/shadiossamamain-rgb/exile-codex

- Shipped a **27 MB PyInstaller build that boots in 3 seconds**, after weighing and rejecting three alternative stacks because an existing CSS design system would not survive a widget toolkit.
- Cut interaction latency from **32.7 ms to 0.7 ms (47×)** by painting in place instead of re-rendering, repairing observer leaks that grew for the lifetime of the process, and coalescing layout reads to one per frame.
- Reached **WCAG AA across 228 text roles**, six screens and both themes, surfacing **six contrast failures** that three narrower prior sweeps had missed; and diagnosed **five host-level failure modes** — including a dialog call that hung the app outright (Windows Application Hang 1002) — with a probe harness logging before and after every risky call.

Path of Exile Data Toolchain — 24 Python modules

2026

Solo build · live trade-API integration, automated validation

- Generate every game search filter **from a live API rather than by hand**, after a patch silently broke the community's hand-maintained filter with no visible symptom — automated re-derivation being the only defence against a failure that produces no error.
- Identified a **14× pricing error** in a naive market-sampling method and replaced it with band-stratified sampling, after proving the naive approach failed silently as a confidently wrong number.

Jarvis — AI-maintained knowledge base & automation suite

2026

Solo build · Python, git, knowledge graph

- Built a **128-page knowledge base with 40 decision records**, a queryable knowledge graph, and **six Python automation scripts** — a daily briefing, a job-board scraper, a video-analysis tool using the Gemini Files API, and three site generators.
- Delivered **two Roblox games across 46 build sessions**, maintaining live project trackers, a 200-level difficulty generator, and an authoritative server architecture for all game state.

SKILLS

AI / LLM: Model Context Protocol (MCP) · Claude Code (skills, hooks, subagents, slash commands) · LLM evaluation & rubric design · Comparative model evaluation · Prompt design · Context management · Gemini API · OpenRouter

Programming: Python · JavaScript · HTML/CSS · Lua · Git · REST APIs · pywebview / WebView2 · PyInstaller · JSON data design

Tools: Blender · Roblox Studio · VS Code · Browser automation · Windows automation

EDUCATION & EARLIER EXPERIENCE

Bachelor's Degree — Geographic Information Systems (GIS), Egyptian Chinese University, Cairo, Egypt

2026 – Expected 2029

High School Diploma — Al Nahda National Private School

Call Center Agent, Etihad Industries · **Seating Attendant**, Yas Circuit Marina · **Traffic Usher**, Bike Abu Dhabi Gran Fondo

2022 – 2023